

EDUCATION

University College London

Studying for BSc Computer Science

2025 to 2028

London, United Kingdom

Key Modules: *Principles of Programming, Algorithms, Object-Oriented Programming*

Garden International School

A-levels: 3A (Computer Science, Mathematics, Further Mathematics)*

2023 to 2025

Kuala Lumpur, Malaysia

AS-levels: A (Physics)

Jerudong International School

IGCSEs: 5A (Mathematics, Computer Science, Physics, Chemistry, Biology)*

2020 to 2022

Bandar Seri Begawan, Brunei

4A (English, English Literature, Additional Mathematics, Design & Technology)

SKILLS

Programming Languages

Python, TypeScript, JavaScript, HTML/CSS, C, Java, Haskell

Technologies

MySQL, NoSQL (DynamoDB, MongoDB), Git, Jira, Confluence

WORK EXPERIENCE

Superstring

Software Engineering Intern

Full-Time | 06/2024 to 07/2024

Kuala Lumpur, Malaysia

- **About the company:** An early-stage **startup** developing **middleware** for businesses to **process and file** government-mandated **e-invoices** with **Malaysia's tax authority (LHDN)**.
- **Developed** dynamic **front-end components** for document management using **React** and **Tailwind CSS**, enhancing **user experience** and enabling efficient document handling.
- **Implemented and integrated RESTful APIs using Node.js and TypeScript** for file upload and download functionalities, enabling seamless data transfer and efficient server communication.
- Integrated mock data generation using **faker.js**, allowing comprehensive testing and improving the reliability of the application.
- Created detailed upload history and validation views with **React** and **DynamoDB**, improving data transparency and reducing error rates during data processing.

[Jewel Digital Ventures](#) (digital subsidiary of Permodalan Nasional Berhad (PNB))

Software Engineering Intern

Full-Time | 02/2023 to 05/2023

Kuala Lumpur, Malaysia

- Developed a **Python** program to filter 8 million emails for duplicates, validate their authenticity, and remove role-based addresses, reinforcing data integrity and compliance with email marketing standards.
- Integrated the cleaned email list into **Sendy (SES)** to enable a targeted campaign for Permodalan Nasional Berhad, optimising outreach and customer engagement.

PROJECTS

Tetris Auto Player,

- Implemented a Tetris auto player with a custom **heuristic model** (holes, bumpiness, height, wells, transitions, line clears) and **lookahead search**.
- Designed logic for **rotation/movement sequences** and intelligent use of bombs/discards based on **predicted board evaluation**.

pH Control and Monitoring System,

- Built an **embedded control system** in **C++ (Arduino)** for real-time pH monitoring and automated actuation.
- Implemented **sensor data processing** using a 10-sample rolling buffer with **trimmed-mean outlier rejection** to reduce noise.
- Developed an **interactive calibration pipeline** via **serial I/O**, applying **linear regression** over 3 reference buffers.
- Designed **setpoint-based control logic (pH 5.0 ± 0.4)** and logged **time-stamped telemetry** for debugging and validation.

Robot Navigation & Collection Simulator,

- Developed a **robot simulator** with autonomous **navigation**, **obstacle avoidance**, and **marker collection** in a **randomised grid**.
- Implemented real-time **graphical visualisation** and an efficient **exploration algorithm** prioritising **least-visited cells**.

POSITIONS OF RESPONSIBILITY

Course Representative for BSc Computer Science Year 1,

Elected by peers to gather student feedback, represent academic interests, and work with staff to improve course experience.

First Year Vice Organising Chairperson (UCL Malaysian Society),

Newly appointed to oversee events and publicity activities, support the Organising Chairperson, and assist in event delivery when required.

Dragon's Coding Club Captain,

Leader of Garden International School's coding club, including organising the annual inter-house coding competition.

Project Eureka - Lead Coding Teacher,

Introduced refugees to technology and coding through Minecraft Education Edition, leading and coordinating volunteer instructors.

Girls In Code SEA Coding Competition Problem Setter and Marker,

Organised and graded coding challenges for an international all-girls coding competition, to promote gender diversity in programming.

AWARDS

PYDC's Game Development Competition Finalist 2023,

Invited to present in Penang for placing top 10 in the country.

Pearson Excellence IGCSE Award 2022,

For achieving excellent results across multiple subjects.